

Sourav Lahon

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EDUCATION

National Institute of Technology Silchar

Bachelor of Technology in Computer Science and Engineering

Cachar, Assam, India

2021 – 2025

EXPERIENCE

BlackRock - Software Engineer | July 2025 – Present

- Maintaining and developing tools to process and transform raw financial data into client-ready deliverables, performing risk-return analysis to ensure accurate and consistent market data, leveraging Perl, Python, SQL.
- Developing scripts to automate the index creation process, reducing creation time from 30 minutes to 5 minutes, with hands-on understanding of index types including fixed income, equity, composites, and hedge indexes.
- Building a real-time data ingestion pipeline using C#, .NET, and Kafka event buses, applying NLP techniques to extract and structure financial information from unstructured data sources.

BlackRock - Data Analyst Intern | Jan 2025 - June 2025

- Built an end-to-end automation application using Python, Pandas, Unix, and Streamlit that generates and deploys configuration files to remote hosts via a custom algorithm, reducing deployment time from 4-5 minutes to under 30 seconds (90% reduction).
- Developed automated reporting scripts in Python and SQL to assess quality control metrics and statuses across market indexes, including database querying, custom algorithm design for quality checks, and structured report generation for operational decision-making, reducing 80% manual effort.

Microsoft - Software Engineer Intern | May 2024 - July 2024

[certificate](#)

- Created a debug tool for the MSAI-UX team, resulting in a 30% reduction in triage time.
- Engineered features including Response Debugger, Accessibility Debugger, and Performance Analyzer, contributing to a 50% performance increase.

PROJECTS

Assam Dress | MongoDB, Express, React, Node

[website](#)

- Built an e-commerce site to promote Assamese culture, reaching 100+ visitors in the first month.
- Implemented secure user authentication with OTP, password hashing, and JWT recovery, reducing login errors by 40% during testing.
- Implemented CRUD operations with Axios and Redux, creating a fluid user interface that streamlined data management for 20+ users during the testing phase.
- Integrated Cloudinary for storing 200+ product images and Stripe for payment processing.
- Integrated Redis for session management, caching frequently accessed product data, cart storage, and rate limiting, improving API response times and system reliability under concurrent user load.

Audio-Visual Speech Recognition (AVSR) | Machine Learning, Python

[thesis](#)

- Designed a multi-modal speech recognition system improving recognition accuracy by 25% in noisy environments compared to traditional audio-only systems.
- Fused audio and video streams using MFCCs and CNNs, resulting in a 20% boost in speech recognition accuracy under noisy conditions.
- Engineered and evaluated classifiers including SVM, Random Forest, DNN, and LSTM, achieving a 30% improvement in recognition rates in challenging acoustic settings.
- Refined feature extraction and data fusion techniques, improving model robustness by 15% and enabling real-time interaction with under 500ms latency in noisy environments.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, JavaScript, Perl, SQL, HTML, CSS

Frameworks & Libraries: React, Redux, React Native, .NET, Node.js, Express.js, NestJS, Flutter, Pandas, Streamlit

Developer Tools: Git, Unix, Kafka, Docker, Redis, MongoDB, Stripe, Cloudinary, Power BI

Concepts: Data Structures, Algorithms, OOP, REST APIs, NLP, Machine Learning, Deep Learning, Event-Driven Architecture, JWT Authentication, Agile/Scrum

CERTIFICATES

Career Essentials in Data Analysis

Career Essentials in Generative AI

Career Essentials in GitHub Professional Certificate

Python Developer Certification by BlackRock

Certificate by Microsoft

Certificate by Microsoft

Certificate by GitHub